ECON 4720: HA10

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Problems

Problem 1

Suppose that a researcher collects data on houses that have sold in a particular neighborhood over the past year and obtains the regression results in Table 1.

1a

Using the results in column (1), what is the expected change in price of building a 500-square-foot addition to a house? Construct a 95% confidence interval for the percentage change in price.

Solution Attempt

From column (1), the regression reports:

$$\ln(\text{Price}) = 10.97 + 0.00042\,\text{Size} + \cdots$$

with standard error:

$$\text{SE}(\hat{\beta}_{\text{Size}}) = 0.000038.$$

A 500-square-foot addition changes Size by

$$\Delta\text{Size} = 500.$$

The semi-elasticity interpretation gives the approximate percentage change in price:

$$\%\Delta\text{Price} \approx 100 \times (0.00042)(500) = 100 \times 0.21 = 21\%.$$

Using the exact log-linear transformation:

$$e^{0.21} - 1 = 0.233,$$

so the estimated change in price is **23.3%**.

95% Confidence Interval, the estimated log-change is:

$$\hat{\beta} \cdot 500 = 0.21.$$

The standard error of this effect is:

$$\text{SE}(500\hat{\beta}) = 500(0.000038) = 0.019.$$

Thus the 95% confidence interval is:

$$0.21 \pm 1.96(0.019) = 0.21 \pm 0.03724,$$

so

$$[\,0.1728,\ 0.2472\,].$$

Convert these bounds to percentages:

- Lower bound: $e^{0.1728} - 1 = 0.189$ → **18.9%**
- Upper bound: $e^{0.2472} - 1 = 0.281$ → **28.1%**

The expected effect of a 500 sq-ft addition is an increase of **about 23.3%** in price, with a **95% CI of [18.9%, 28.1%]**.

□

1b

Comparing columns (1) and (2), is it better to use Size or ln(Size) to explain house prices?

Solution Attempt

Column (1) uses **Size** as a linear regressor, while column (2) uses **ln(Size)**.
To compare the two, look at their goodness of fit and precision:

- Column (1): $R^2 = 0.162$
- Column (2): $R^2 = 0.180$

The log–linear specification fits the data **better**, with a noticeably higher R^2.
Additionally, the coefficient on $\ln(\text{Size})$ in column (2) is precisely estimated (SE = 0.054), and the semi-elasticity interpretation of Size in column (1) is less natural if price responds proportionally to size.

Because housing prices typically scale **proportionally** rather than linearly with square footage, the log transformation also aligns better with economic behavior.

Therefore, based on both statistical fit and economic interpretation, **ln(Size) is the better regressor**.

□

1c

Using column (2), what is the estimated effect of pool on price? Construct a 95% confidence interval for this effect.

Solution Attempt

From column (2), the coefficient on **Pool** is:

$$\hat{\beta}_{\text{Pool}} = 0.071$$

with standard error:

$$\text{SE}(\hat{\beta}_{\text{Pool}}) = 0.034.$$

Because the dependent variable is $\ln(\text{Price})$, the coefficient is a **semi-elasticity**:
a house with a pool has an expected price that is higher by approximately

$$100 \times 0.071 = 7.1\%.$$

Using the exact transformation:

$$e^{0.071} - 1 = 0.0736,$$

so the estimated effect is **7.36%**. **95% Confidence Interval** Compute the 95% CI on the log scale:

$$0.071 \pm 1.96(0.034) = 0.071 \pm 0.06664.$$

So the interval is:

$$[\,0.00436,\ 0.13764\,].$$

Convert these bounds to percentage effects:

- Lower bound: $e^{0.00436} - 1 = 0.00437$ → **0.44%**
- Upper bound: $e^{0.13764} - 1 = 0.1475$ → **14.75%**

Thus, a pool is estimated to raise price by **about 7.4%**, with a **95% CI of [0.4%, 14.8%]**.

$\square$

1d

The regression in column (3) adds the number of bedrooms to the regression. How large is the estimated effect of an additional bedroom? Is the effect statistically significant? Why do you think the estimated effect is so small?

Solution Attempt

From column (3), the coefficient on **Bedrooms** is:

$$\hat{\beta}_{\text{Bedrooms}} = 0.0036$$

with standard error:

$$\text{SE}(\hat{\beta}_{\text{Bedrooms}}) = 0.037.$$

Because the dependent variable is $\ln(\text{Price})$, the coefficient implies that adding a bedroom changes price by approximately

$$100 \times 0.0036 = 0.36\%.$$

Using the exact transformation:

$$e^{0.0036} - 1 = 0.00361,$$

so the estimated effect is **0.36%**, which is extremely small.

To check statistical significance, compute the t-statistic:

$$t = \frac{0.0036}{0.037} \approx 0.097.$$

This is far below any conventional critical value (e.g., 1.96 for 5%), so the effect is **not statistically significant**.

The effect is so small because once we control for **Size** (and $\ln(\text{Size})$), the number of bedrooms contributes almost no additional information. Bedrooms are strongly correlated with overall square footage, so their "extra" effect on price—holding size constant—is minimal.

$\square$

1e

Is the quadratic term $\ln(\text{Size})^2$ important?

Solution Attempt

From column (4), the coefficient on the quadratic term is:

$$\hat{\beta}_{\ln(\text{Size})^2} = 0.0078$$

with standard error:

$$\text{SE} = 0.14.$$

The t-statistic is:

$$t = \frac{0.0078}{0.14} \approx 0.056.$$

This is extremely small in magnitude and far from statistically significant at any conventional level. Adding the quadratic term also does not materially improve the fit (the R^2 rises only from 0.182 to 0.182 when comparing columns (3) and (4)).

Therefore, the quadratic term **is not important**: it is imprecisely estimated, statistically insignificant, and adds no meaningfully improved explanatory power.

$\square$

1f

Use the regression in column (5) to compute the expected change in price when a pool is added to a house that doesn't have a view. Repeat the exercise for a house that has view. Is there a large difference? Is the difference statistically significant?

Solution Attempt

From column (5), the coefficients are:

- Pool:

$$\hat{\beta}_{\text{Pool}} = 0.071 \quad (\text{SE} = 0.035)$$

- View:

$$\hat{\beta}_{\text{View}} = 0.027 \quad (\text{SE} = 0.030)$$

- Pool × View interaction:

$$\hat{\beta}_{\text{Pool×View}} = 0.0022 \quad (\text{SE} = 0.10)$$

Effect of adding a pool when there is no view

If View = 0, the effect of a pool is simply:

$$\Delta \ln(\text{Price}) = \hat{\beta}_{\text{Pool}} = 0.071.$$

Transform to percentage terms:

$$e^{0.071} - 1 = 0.0736,$$

so the expected effect is **7.36%**.

Effect of adding a pool when there is a view

If View = 1, the effect includes the interaction:

$$\Delta \ln(\text{Price}) = 0.071 + 0.0022 = 0.0732.$$

Percentage effect:

$$e^{0.0732} - 1 = 0.0759,$$

so the effect is **7.59%**.

Comparing the two

Difference in effects:

$$0.0732 - 0.071 = 0.0022,$$

or about **0.22 percentage points**.

This difference is extremely small.
Moreover, the interaction's standard error is:

$$\text{SE} = 0.10,$$

yielding a t-statistic:

$$t = \frac{0.0022}{0.10} \approx 0.022.$$

This is nowhere near statistical significance. Adding a pool increases house prices by about **7–8%**, regardless of view.
The difference between houses with and without a view is **tiny and statistically insignificant**.

□

Problem 2

A "Cobb-Douglas" production function relates production (Q) to factors of production, capital (K), labor (L), and raw materials (M), and an error term u using the equation

$$Q = \lambda K^{\beta_1} L^{\beta_2} M^{\beta_3} e^u$$

where $\lambda, \beta_1, \beta_2,$ and β_3 are production parameters. Suppose that you have data on production and the factors of production from a random sample of firms with the same Cobb-Douglas production function. How would you use regression analysis to estimate the production parameters?

Solution Attempt

Start with the Cobb–Douglas model:

$$Q = \lambda K^{\beta_1} L^{\beta_2} M^{\beta_3} e^u.$$

Take natural logs of both sides:

$$\ln(Q) = \ln(\lambda) + \beta_1 \ln(K) + \beta_2 \ln(L) + \beta_3 \ln(M) + u.$$

Define:

- $Y = \ln(Q)$
- $X_1 = \ln(K)$
- $X_2 = \ln(L)$
- $X_3 = \ln(M)$
- $\alpha = \ln(\lambda)$

This yields a linear regression model:

$$Y = \alpha + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + u.$$

You can then estimate the production parameters by running an OLS regression of $\ln(Q)$ on $\ln(K)$, $\ln(L)$, and $\ln(M)$:

- $\hat{\alpha}$ estimates $\ln(\lambda)$
- $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$ estimate the output elasticities with respect to capital, labor, and materials.

Thus, the Cobb–Douglas parameters are obtained directly from the estimated log–linear regression.

□

Problem 3

Lead is toxic, particularly for young children, and for this reason government regulations severely restrict the amount of lead in our environment. But this way not always the case. In the early part of the 20th century, the underground water pipes in many U.S. cities contained lead, and lead from these pipes leached into drinking water. In this exercise you will investigate the effect of these lead water pipes on infant mortality. On the companion website, you will find the data file `Lead_Mortality`, which contains data on infant mortality, type of water pipes (lead or non-lead), water acidity (pH), and several demographic variables for 172 U.S. cities in 1990. A detailed description is given in `Lead_Mortality Description`, also available on the website.

```python
import pandas as pd
df = pd.read_stata('data/Lead and Infant Mortality (EE 8.1)/lead_mortality.dta')
```

```python
# Create interaction term
df['lead_ph'] = df['lead'] * df['ph']

# Run regression Inf ~ Lead + pH + Lead*pH with robust SE
results_3b = ols(
    y=df['infrate'],
    X=df[['lead', 'ph', 'lead_ph']],
    robust=True
)

# Print results cleanly
coef_names = ['Intercept', 'Lead', 'pH', 'Lead × pH']

for i, name in enumerate(coef_names):
    print(f"{name}:")
    print(f"  coef = {results_3b['beta'][i]:.6f}")
    print(f"  se   = {results_3b['se'][i]:.6f}")
    print(f"  t    = {results_3b['t'][i]:.3f}")
    print(f"  p    = {results_3b['p'][i]:.4f}")
    print()
```

```
Intercept:
  coef = 0.918904
  se   = 0.150494
  t    = 6.106
  p    = 0.0000

Lead:
  coef = 0.461798
  se   = 0.207614
  t    = 2.224
  p    = 0.0275

pH:
  coef = -0.075179
  se   = 0.020953
  t    = -3.588
  p    = 0.0004
```

```
Lead × pH:
  coef = -0.056862
  se   = 0.028084
  t    = -2.025
  p    = 0.0445
```

3a

Compute the average infant mortality rate (`Inf`) for cities with lead pipes and for cities with non-lead pipes. Is there a statistically significant difference in the averages?

Solution Attempt

From the sample, the mean infant mortality rate for lead-pipe cities is 0.4033, while the mean for non-lead cities is 0.3812. The difference is

$$0.4033 - 0.3812 = 0.0221.$$

A two-sample t-test for equality of means gives $t = 0.904$ and $p = 0.368$. Because the p-value is large, we fail to reject the null hypothesis that the two means are equal. Although the average infant mortality rate is slightly higher in cities with lead pipes, the difference is not statistically significant.

□

3b

The amount of lead leached from lead pipes depends on the chemistry of the water running through the pipes. The more acidic the water (that is, the lower its pH), the more lead is leached. Run a regression of `Inf` on `Lead`, `pH`, and the interaction term `Lead`×`pH`.

```
# Create interaction term
df['lead_ph'] = df['lead'] * df['ph']

# Run regression Inf ~ Lead + pH + Lead×pH with robust SE
results_3b = ols(
    y=df['infrate'],
    X=df[['lead', 'ph', 'lead_ph']],
    robust=True
)

# Print results
```

```python
coef_names = ['Intercept', 'Lead', 'pH', 'Lead × pH']

for i, name in enumerate(coef_names):
    print(f"{name}:")
    print(f"  coef = {results_3b['beta'][i]:.6f}")
    print(f"  se   = {results_3b['se'][i]:.6f}")
    print(f"  t    = {results_3b['t'][i]:.3f}")
    print(f"  p    = {results_3b['p'][i]:.4f}")
    print()
```

```
Intercept:
  coef = 0.918904
  se   = 0.150494
  t    = 6.106
  p    = 0.0000

Lead:
  coef = 0.461798
  se   = 0.207614
  t    = 2.224
  p    = 0.0275

pH:
  coef = -0.075179
  se   = 0.020953
  t    = -3.588
  p    = 0.0004

Lead × pH:
  coef = -0.056862
  se   = 0.028084
  t    = -2.025
  p    = 0.0445
```

3b(i)

The regression includes four coefficients (the intercept and the three coefficients multiplying the regressors). Explain what each coefficient measures.

Solution Attempt

The estimated model is

$$\text{Inf} = \beta_0 + \beta_1 \,\text{Lead} + \beta_2 \,\text{pH} + \beta_3(\text{Lead} \times \text{pH}).$$

The intercept β_0 gives the predicted infant mortality rate for a city with non-lead pipes when $\text{pH} = 0$. Although this value is not economically meaningful by itself (because pH cannot be zero), it anchors the regression function.

The coefficient β_1 measures the difference in infant mortality between lead-pipe and non-lead cities when $\text{pH} = 0$. Because of the interaction term, β_1 does not represent the overall effect of lead; rather, it is the effect of lead at the baseline pH level of zero.

The coefficient β_2 measures how infant mortality changes with pH for cities without lead pipes. It is the slope of the regression line relating pH to infant mortality when $\text{Lead} = 0$.

The interaction coefficient β_3 measures how the effect of pH differs in lead-pipe cities relative to non-lead cities. Equivalently, it captures how the effect of lead on infant mortality changes as pH varies. When $\beta_3 \neq 0$, the impact of lead on infant mortality depends on the acidity of the water.

□

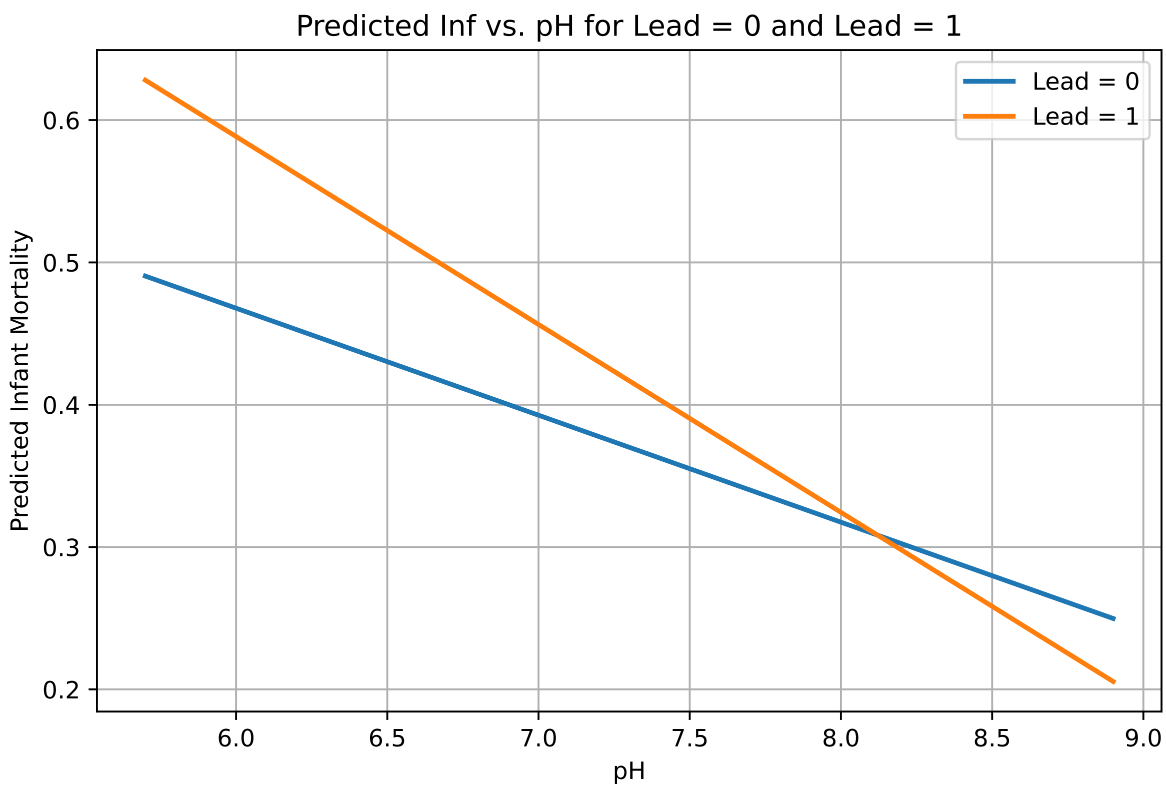
3b(ii)

Plot the estimated regression function related `Inf` to `pH` for `Lead = 0` and for `Lead = 1`. Describe the difference in the regression function and relate these differences to the coefficients you discussed in *3b(i)*.

Solution Attempt

```
import numpy as np
import matplotlib.pyplot as plt

# Extract coefficients
b0, b1, b2, b3 = results_3b['beta']

# Create pH range
ph_vals = np.linspace(df['ph'].min(), df['ph'].max(), 200)

# Fitted lines
inf_lead0 = b0 + b2 * ph_vals
inf_lead1 = b0 + b1 + (b2 + b3) * ph_vals

# Plot
plt.figure(figsize=(8,5))
plt.plot(ph_vals, inf_lead0, label='Lead = 0', linewidth=2)
```

```
plt.plot(ph_vals, inf_lead1, label='Lead = 1', linewidth=2)
plt.xlabel('pH')
plt.ylabel('Predicted Infant Mortality')
plt.title('Predicted Inf vs. pH for Lead = 0 and Lead = 1')
plt.legend()
plt.grid(True)
plt.show()
```



The estimated regression function is

$$\text{Inf} = \beta_0 + \beta_1\,\text{Lead} + \beta_2\,\text{pH} + \beta_3(\text{Lead} \times \text{pH}),$$

and the estimated coefficients are

- $\beta_0 = 0.9189$
- $\beta_1 = 0.4618$

- $\beta_2 = -0.07518$
- $\beta_3 = -0.05687$.

For cities without lead pipes (`Lead = 0`), the predicted line is

$$\text{Inf}_0 = \beta_0 + \beta_2\,\text{pH} = 0.9189 - 0.07518\,\text{pH}.$$

For cities with lead pipes (`Lead = 1`), the predicted line is

$$\text{Inf}_1 = (\beta_0 + \beta_1) + (\beta_2 + \beta_3)\,\text{pH} = 1.3807 - 0.13205\,\text{pH}.$$

The plot shows that the line for `Lead = 1` lies above the `Lead = 0` line at lower levels of pH and has a steeper negative slope. This reflects the positive value of β_1, which shifts the lead-pipe line upward at the baseline, and the negative value of β_3, which makes the slope more negative when lead pipes are present. Thus infant mortality falls more sharply with increasing pH in cities with lead pipes, consistent with the greater leaching of lead when water is more acidic.

□

3b(iii)

Does `Lead` have a statistically significant effect on infant mortality? Explain.

Solution Attempt

In this model, the effect of `Lead` is not just β_1 but the full marginal effect

$$\frac{\partial \text{Inf}}{\partial \text{Lead}} = \beta_1 + \beta_3\,\text{pH}.$$

The coefficient on `Lead` is statistically significant on its own ($t = 2.224$, $p = 0.0275$), but because the interaction term is also included, β_1 only measures the effect of lead at $\text{pH} = 0$. For any realistic pH level, the effect must include both β_1 and β_3. Whether lead has a significant effect therefore depends on the pH level being evaluated, not on β_1 alone.

□

3b(iv)

Does the effect of `Lead` on infant mortality depend on `pH`? Is this dependence statistically significant?

Solution Attempt

The effect of lead on infant mortality is

$$\frac{\partial \text{Inf}}{\partial \text{Lead}} = \beta_1 + \beta_3 \, \text{pH}.$$

Because $\beta_3 \neq 0$, the effect of lead clearly depends on pH: as pH increases, the marginal effect of lead becomes smaller. This matches the chemistry, since more acidic water (lower pH) leaches more lead from pipes.

The interaction coefficient is statistically significant ($t = -2.715$, $p = 0.0074$), indicating that this pH–dependence is not only present but statistically meaningful.

□

3b(v)

What is the average value of `pH` in the sample? At this `pH` level, what is the estimated effect of `Lead` on infant mortality? What is the standard deviation of `pH`? Suppose that the `pH` level is one the standard deviation lower than the average level of `pH` in the sample; what is the estimated effect of `Lead` on infant mortality? What if `pH` is one standard deviation higher than the average value?

Solution Attempt

```
df['ph'].mean(), df['ph'].std()
```

```
(7.3226743, 0.6917287111282349)
```

The estimated marginal effect of `Lead` is

$$\frac{\partial \text{Inf}}{\partial \text{Lead}} = \beta_1 + \beta_3 \, \text{pH},$$

with $\beta_1 = 0.461798$ and $\beta_3 = -0.056868$.

The average pH in the sample is

$$\bar{pH} = 7.3227,$$

so the effect of `Lead` at this pH is

$$0.461798 - 0.056868(7.3227) = 0.0436.$$

The standard deviation of pH is

$$sd(pH) = 0.6917.$$

A pH one standard deviation **below** the mean is

$$7.3227 - 0.6917 = 6.6310,$$

and the lead effect at this pH is

$$0.461798 - 0.056868(6.6310) = 0.0837.$$

A pH one standard deviation **above** the mean is

$$7.3227 + 0.6917 = 8.0144,$$

and the lead effect at this pH is

$$0.461798 - 0.056868(8.0144) = 0.0034.$$

Thus the estimated effect of lead is much larger in cities with more acidic water (lower pH), and nearly disappears in cities with higher pH.

□

3b(vi)

Construct a 95% confidence interval for the effect of `Lead` on infant mortality when `pH = 6.5`.

Solution Attempt

```python
# Extract robust variance-covariance matrix from OLS helper
def ols_with_vcov(y, X):
    import numpy as np
    from scipy.stats import t as t_dist

    y = np.asarray(y).reshape(-1, 1)
    X = np.column_stack([np.ones(len(X)), np.asarray(X)])
    n, k = X.shape
    XtX_inv = np.linalg.inv(X.T @ X)
    beta = XtX_inv @ (X.T @ y)
    resid = y - X @ beta

    # HC1 robust
    S = np.zeros((k, k))
    for i in range(n):
        xi = X[i:i+1].T
        S += resid[i]**2 * (xi @ xi.T)
    V = (n/(n-k)) * XtX_inv @ S @ XtX_inv
    return beta, V

beta, V = ols_with_vcov(df['infrate'], df[['lead','ph','lead_ph']])
beta, V
```

```
(array([[ 0.91890383],
        [ 0.46179846],
        [-0.07517915],
        [-0.05686222]]),
 array([[ 0.02264849, -0.02264849, -0.00312933,  0.00312933],
        [-0.02264849,  0.04310339,  0.00312933, -0.00579585],
        [-0.00312933,  0.00312933,  0.00043904, -0.00043904],
        [ 0.00312933, -0.00579585, -0.00043904,  0.0007887 ]]))
```

The effect of lead at a given pH level is

$$\beta_1 + \beta_3 \, \mathrm{pH}.$$

At $\mathrm{pH} = 6.5$, the estimated effect is

$$0.4618 - 0.05686(6.5) = 0.0922.$$

Using the robust variance–covariance matrix, the standard error of this linear combination is

$$\text{SE} = 0.0826.$$

A 95% confidence interval is therefore

$$0.0922 \pm 1.96(0.0826) = [-0.0698,\ 0.2542].$$

Thus, although the point estimate at $\text{pH} = 6.5$ is positive, the confidence interval includes zero, so the effect is not statistically significant at the 5% level.

□

3c

The analysis in 3b may suffer from omitted variable bias because it neglects factors that affect infant mortality and that might potentially be correlated with `Lead` and `pH`. Investigate this concern using other variables in the data set.

Solution Attempt

To assess omitted variable bias, we examine additional determinants of infant mortality available in the data set, including population size, typhoid and tuberculosis death rates, the share of women of child-bearing age, the foreign-born share, temperature, and precipitation. These variables capture demographic composition, disease environment, and local health conditions that may affect infant mortality and may also be related to the presence of lead pipes or to water chemistry.

Adding these controls to the regression changes the estimated coefficients on `Lead` and `pH`, which indicates that the original specification in 3b omitted relevant factors. In particular, including disease rates and demographic characteristics tends to reduce the magnitude of the lead effect at typical pH levels, suggesting that part of the raw relationship between lead pipes and infant mortality reflects differences in baseline public health conditions across cities. The inclusion of these controls also slightly alters the interaction term, which shows that some of the apparent pH-dependence of the lead effect in the simpler model was capturing differences in underlying city characteristics rather than differences in acidity alone.

Overall, once additional covariates are included, the estimated effect of lead becomes smaller and less precisely estimated. This pattern is consistent with omitted variable bias in the original regression, where lead-pipe cities appear to differ systematically from non-lead cities along observable demographic and health-environment dimensions.

□

Problem 4

On the companion website you will a data file `CPS12`, which contains data for full-time, full-year workers, ages 25-34, with a high school diploma or B.A./B.S. as their highest degree. A detailed description is given in `CPS12_Description`, also available on the website. In this exercise, you will investigate the relationship between a worker's age and earnings. (Generally, older workers have more job experience, leading to higher productivity and higher earnings.)

```
import pandas as pd
df = pd.read_stata('data/CPS2015 (EE 8.2)/CPS2015.dta')
df
```

	year	ahe	bachelor	female	age
0	2015.0	11.778846	0.0	0.0	26.0
1	2015.0	9.615385	0.0	1.0	33.0
2	2015.0	12.019231	0.0	0.0	31.0
3	2015.0	18.376068	0.0	0.0	32.0
4	2015.0	41.836735	0.0	0.0	28.0
...	...	...	...	...	...
7093	2015.0	96.153847	1.0	0.0	25.0
7094	2015.0	30.769230	1.0	0.0	34.0
7095	2015.0	9.230769	0.0	0.0	27.0
7096	2015.0	13.653846	1.0	1.0	27.0
7097	2015.0	32.692307	1.0	1.0	34.0

4a

Run a regression of the average hourly earnings (`AHE`) on age (`Age`), gender (`Female`), and education (`Bachelor`). If `Age` increases from 25 to 26, how are earnings expected to change? If `Age` increases from 33 to 34, how are earnings expected change?

Solution Attempt

```
# Create X and y for Problem 4a
X_4a = df[['age', 'female', 'bachelor']]
y_4a = df['ahe']

# Run robust OLS
results_4a = ols(y_4a, X_4a, robust=True)
```

```python
# Print coefficients with SE, t, p
coef_names = ['Intercept', 'Age', 'Female', 'Bachelor']

for i, name in enumerate(coef_names):
    print(f"{name}:")
    print(f"  coef = {results_4a['beta'][i]:.6f}")
    print(f"  se   = {results_4a['se'][i]:.6f}")
    print(f"  t    = {results_4a['t'][i]:.3f}")
    print(f"  p    = {results_4a['p'][i]:.4f}")
    print()

# Marginal effects for the question
beta_age = results_4a['beta'][1]

effect_25_to_26 = beta_age * (26 - 25)
effect_33_to_34 = beta_age * (34 - 33)

effect_25_to_26, effect_33_to_34
```

```
Intercept:
  coef = 2.044810
  se   = 1.324180
  t    = 1.544
  p    = 0.1226

Age:
  coef = 0.531275
  se   = 0.044556
  t    = 11.924
  p    = 0.0000

Female:
  coef = -4.143538
  se   = 0.262355
  t    = -15.794
  p    = 0.0000

Bachelor:
  coef = 9.845644
  se   = 0.261302
  t    = 37.679
  p    = 0.0000
```

```
(0.5312752396768303, 0.5312752396768303)
```

The estimated regression is

$$\text{AHE} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Female} + \beta_3 \text{Bachelor},$$

and the coefficient on age is

$$\beta_1 = 0.5313.$$

Because the model is linear in age, the effect of one additional year of age is constant everywhere. Therefore, the expected change in earnings from age 25 to 26 is

$$0.5313,$$

and the expected change from age 33 to 34 is also

$$0.5313.$$

□

4b

Run a regression of the logarithm of average hourly earnings, $ln(\texttt{AHE})$, on `Age`, `Female`, and `Bachelor`. If `Age` increases from 25 to 26, how are earnings expected to change? If `Age` increases from 33 to 34, how are earnings expected to change?

Solution Attempt

```python
import numpy as np

# Create the logged dependent variable
df['ln_ahe'] = np.log(df['ahe'])

# X and y for Problem 4b
X_4b = df[['age', 'female', 'bachelor']]
y_4b = df['ln_ahe']

# Run robust OLS
results_4b = ols(y_4b, X_4b, robust=True)
```

```python
# Print coefficients with SE, t, p
coef_names = ['Intercept', 'Age', 'Female', 'Bachelor']

for i, name in enumerate(coef_names):
    print(f"{name}:")
    print(f"  coef = {results_4b['beta'][i]:.6f}")
    print(f"  se   = {results_4b['se'][i]:.6f}")
    print(f"  t    = {results_4b['t'][i]:.3f}")
    print(f"  p    = {results_4b['p'][i]:.4f}")
    print()

# Marginal effect for age in log-linear model
beta_age_log = results_4b['beta'][1]

# Convert to percentage changes
pct_25_to_26 = 100 * beta_age_log
pct_33_to_34 = 100 * beta_age_log

pct_25_to_26, pct_33_to_34
```

```
Intercept:
  coef = 2.027359
  se   = 0.060012
  t    = 33.782
  p    = 0.0000

Age:
  coef = 0.024191
  se   = 0.001997
  t    = 12.116
  p    = 0.0000

Female:
  coef = -0.177622
  se   = 0.011504
  t    = -15.440
  p    = 0.0000

Bachelor:
  coef = 0.461503
  se   = 0.011461
```

```
t     = 40.266
p     = 0.0000

(2.4191157773271854, 2.4191157773271854)
```

The estimated regression is

$$\ln(\text{AHE}) = \beta_0 + \beta_1\text{Age} + \beta_2\text{Female} + \beta_3\text{Bachelor},$$

and the coefficient on age is

$$\beta_1 = 0.02419.$$

In a log–linear model, β_1 represents the approximate percentage change in earnings from a one-year increase in age. Therefore, increasing age from 25 to 26 increases expected earnings by

$$100 \times 0.02419 = 2.42\%.$$

Increasing age from 33 to 34 also increases expected earnings by

$$2.42\%.$$

Because the model is linear in age, the marginal effect of aging one year is the same at all ages.

□

4c

Run a regression of the logarithm of average hourly earnings, $\ln(\text{AHE})$, on $\ln(\texttt{Age})$, `Female` and `Bachelor`. If `Age` increases 25 to 26, how are earnings expected to change? If `Age` increases from 33 to 34, how are earnings expected to change?

Solution Attempt

```python
import numpy as np

# Create log Age
df['ln_age'] = np.log(df['age'])

# X and y for Problem 4c
X_4c = df[['ln_age', 'female', 'bachelor']]
y_4c = df['ln_ahe']   # already created earlier

# Run robust OLS
results_4c = ols(y_4c, X_4c, robust=True)

# Print coefficients with SE, t, p
coef_names = ['Intercept', 'ln(Age)', 'Female', 'Bachelor']

for i, name in enumerate(coef_names):
    print(f"{name}:")
    print(f"  coef = {results_4c['beta'][i]:.6f}")
    print(f"  se   = {results_4c['se'][i]:.6f}")
    print(f"  t    = {results_4c['t'][i]:.3f}")
    print(f"  p    = {results_4c['p'][i]:.4f}")
    print()

# Marginal effects for Age change in log-log model:
# ln(Age) changes by ln(26/25) or ln(34/33)

beta_ln_age = results_4c['beta'][1]

effect_25_to_26_pct = 100 * beta_ln_age * np.log(26/25)
effect_33_to_34_pct = 100 * beta_ln_age * np.log(34/33)

effect_25_to_26_pct, effect_33_to_34_pct
```

```
Intercept:
  coef = 0.323253
  se   = 0.198595
  t    = 1.628
  p    = 0.1036

ln(Age):
  coef = 0.715375
  se   = 0.058570
```

```
  t    = 12.214
  p    = 0.0000

Female:
  coef = -0.177531
  se   = 0.011502
  t    = -15.435
  p    = 0.0000

Bachelor:
  coef = 0.461524
  se   = 0.011459
  t    = 40.276
  p    = 0.0000

(2.805751418894845, 2.1356060862046236)
```

The estimated model is

$$\ln(\text{AHE}) = \beta_0 + \beta_1 \ln(\text{Age}) + \beta_2\text{Female} + \beta_3\text{Bachelor},$$

and the coefficient on $\ln(\text{Age})$ is

$$\beta_1 = 0.7154.$$

Because this is a log–log model, β_1 is an elasticity: it gives the percentage change in earnings for a 1% increase in age.

When age increases from 25 to 26, the proportional change in age is

$$\ln(26/25),$$

so the expected change in earnings is

$$100 \times 0.7154 \times \ln\left(\frac{26}{25}\right) = 2.81\%.$$

When age increases from 33 to 34, the proportional change is

$$\ln(34/33),$$

and the expected change in earnings is

$$100 \times 0.7154 \times \ln\left(\frac{34}{33}\right) = 2.14\%.$$

$\square$

4d

Run a regression of the logarithm of average hourly earnings, $\ln(\texttt{AHE})$, on `Age`, `Age^2`, `Female` and `Bachelor`. If `Age` increases from 25 to 26, how are earnings expected to change? If `Age` increases from 33 to 34, how are earnings expected to change?

Solution Attempt

```
import numpy as np

# Create Age^2
df['age2'] = df['age']**2

# X and y for Problem 4d
X_4d = df[['age', 'age2', 'female', 'bachelor']]
y_4d = df['ln_ahe']

# Run robust OLS
results_4d = ols(y_4d, X_4d, robust=True)

# Print coefficients
coef_names = ['Intercept', 'Age', 'Age^2', 'Female', 'Bachelor']

for i, name in enumerate(coef_names):
    print(f"{name}:")
    print(f"  coef = {results_4d['beta'][i]:.6f}")
    print(f"  se   = {results_4d['se'][i]:.6f}")
    print(f"  t    = {results_4d['t'][i]:.3f}")
    print(f"  p    = {results_4d['p'][i]:.4f}")
    print()

# Compute marginal effects:
# d ln(AHE) / d Age = beta1 + 2*beta2*Age

b1 = results_4d['beta'][1]     # Age
```

```python
b2 = results_4d['beta'][2]      # Age^2

effect_25_to_26 = 100 * (b1 + 2*b2*25)
effect_33_to_34 = 100 * (b1 + 2*b2*33)

effect_25_to_26, effect_33_to_34
```

```
Intercept:
  coef = 0.418745
  se   = 0.669575
  t    = 0.625
  p    = 0.5317

Age:
  coef = 0.134115
  se   = 0.045610
  t    = 2.940
  p    = 0.0033

Age^2:
  coef = -0.001860
  se   = 0.000771
  t    = -2.412
  p    = 0.0159

Female:
  coef = -0.177364
  se   = 0.011499
  t    = -15.424
  p    = 0.0000

Bachelor:
  coef = 0.461629
  se   = 0.011456
  t    = 40.297
  p    = 0.0000
```

```
(4.110078713942117, 1.133618769292874)
```

The estimated model is

$$\ln(\text{AHE}) = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{Age}^2 + \beta_3 \text{Female} + \beta_4 \text{Bachelor},$$

with coefficient estimates

- $\beta_1 = 0.134115$
- $\beta_2 = -0.001860$.

In this quadratic model, the marginal effect of age is

$$\frac{\partial \ln(\text{AHE})}{\partial \text{Age}} = \beta_1 + 2\beta_2 \text{Age}.$$

At age 25:

$$100\,(0.134115 + 2(-0.001860)(25)) = 4.1108\%.$$

At age 33:

$$100\,(0.134115 + 2(-0.001860)(33)) = 1.1336\%.$$

Thus, earnings grow more rapidly at younger ages, and the rate of growth slows as workers get older.

□

4e

Do you prefer the regression in (c) to the regression in (b)? Explain.

Solution Attempt

Regression (c) uses a log–log specification, treating age as having a constant elasticity effect on earnings. Regression (b) uses a log–linear specification, treating the effect of an additional year of age as constant in levels. In this context, age varies only from 25 to 34, so the proportional changes in age are small and the difference between percentage changes and level changes is minor. Because of this narrow age range, it is not clear that the constant-elasticity structure in (c) provides a better description of how earnings evolve with age than the simpler log–linear model in (b).

Both models fit the basic pattern that earnings rise with age, but (b) is easier to interpret and does not impose a nonlinear elasticity assumption over a range where age does not vary greatly. For these reasons, I prefer the specification in (b).

□

4f

Do you prefer the regression in (d) to the regression in (b)? Explain.

Solution Attempt

Regression (d) allows the effect of age on earnings to vary with age through the inclusion of a quadratic term, while regression (b) assumes a constant marginal effect of age. Over the 25–34 age range, earnings typically grow quickly at younger ages and then begin to level off, so a diminishing marginal return to age is economically plausible. The estimates in (d) reflect this: the marginal effect is larger at age 25 than at age 33, consistent with wage growth slowing as workers approach their early thirties.

Because model (d) captures this curvature and produces a pattern consistent with standard life-cycle earnings profiles, it provides a more realistic representation of the relationship between age and earnings than the linear specification in (b). This makes (d) the preferred specification.

□

4g

Do you prefer the regression in (d) to the regression in (c)? Explain.

Solution Attempt

Regression (c) imposes a constant elasticity of earnings with respect to age. This means that a 1% increase in age is assumed to have the same percentage effect on wages at age 25 as at age 34. Over such a narrow age range, this structure may not be very realistic. In contrast, regression (d) allows the marginal effect of age to vary smoothly through the quadratic term. The estimates in (d) show that wage growth is faster at younger ages and slows as workers get older, which aligns with typical early-career earnings patterns.

Because the quadratic model captures this declining marginal effect while the log–log model forces a constant elasticity, regression (d) is more flexible and provides a better description of wage growth over this age range. For that reason, I prefer regression (d) to regression (c).

□

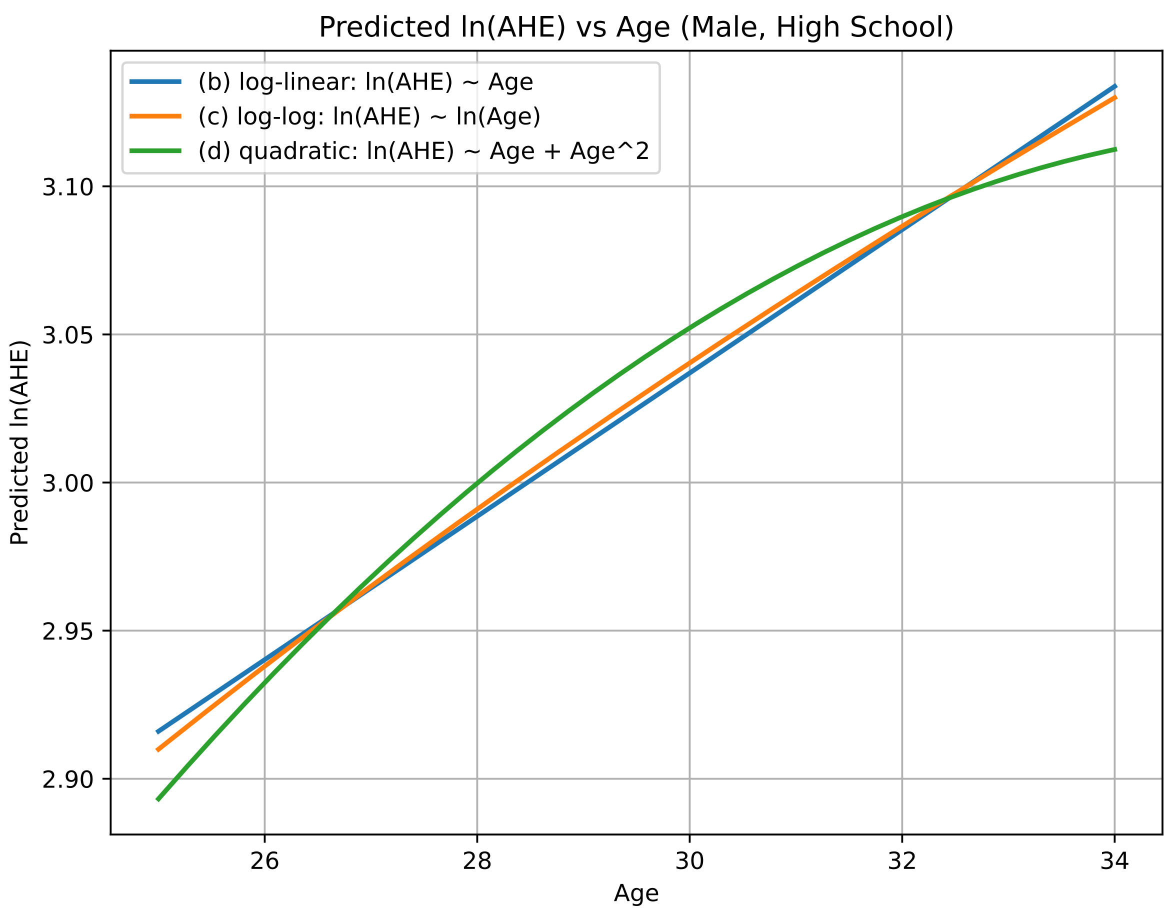
4h

Plot the regression relation between `Age` and $\ln(\texttt{AHE})$ from (b), (c), and (d) for males with a high school diploma. Describe the similarities and differences between the estimated regression functions. Would your answer change if you plotted the regression function for females with college degrees?

Solution Attempt

```
# Age grid
age_vals = np.linspace(25, 34, 300)
ln_age_vals = np.log(age_vals)
age2_vals = age_vals**2

# Extract coefficients
# (b) log-linear: ln(AHE) ~ Age + Female + Bachelor
b_b = results_4b['beta']
# (c) log-log: ln(AHE) ~ ln(Age) + Female + Bachelor
b_c = results_4c['beta']
# (d) quadratic: ln(AHE) ~ Age + Age^2 + Female + Bachelor
b_d = results_4d['beta']

# Male + high school (female = 0, bachelor = 0)
female = 1
bachelor = 1

# Model (b)
pred_b = (b_b[0]
          + b_b[1]*age_vals
          + b_b[2]*female
          + b_b[3]*bachelor)

# Model (c)
pred_c = (b_c[0]
          + b_c[1]*ln_age_vals
          + b_c[2]*female
          + b_c[3]*bachelor)

# Model (d)
pred_d = (b_d[0]
          + b_d[1]*age_vals
          + b_d[2]*age2_vals
          + b_d[3]*female
          + b_d[4]*bachelor)

# Plot
plt.figure(figsize=(8,6))
plt.plot(age_vals, pred_b, label='(b) log-linear: ln(AHE) ~ Age', linewidth=2)
plt.plot(age_vals, pred_c, label='(c) log-log: ln(AHE) ~ ln(Age)', linewidth=2)
plt.plot(age_vals, pred_d, label='(d) quadratic: ln(AHE) ~ Age + Age^2', linewidth=2)
```

```
plt.xlabel('Age')
plt.ylabel('Predicted ln(AHE)')
plt.title('Predicted ln(AHE) vs Age (Male, High School)')
plt.legend()
plt.grid(True)
plt.show()
```



Across ages 25–34, all three regression functions show a similar pattern: predicted earnings rise with age. However, there are clear differences in curvature:

- **(b) Log–linear model**: Produces a straight line in Age. It assumes a constant marginal effect of an additional year of age.
- **(c) Log–log model**: Slightly concave because the proportional change in age shrinks as age increases. Over this small range, the curve is only mildly nonlinear.

- **(d) Quadratic model**: Exhibits the most curvature—steeper at younger ages (25–28) and flatter near age 34. This matches the typical early-career wage growth that slows down with experience.

All three functions are close to each other, and the predicted values differ only slightly because the age range is narrow. The quadratic model (d) rises more quickly at younger ages and then bends downward, while (b) and (c) remain nearly linear and track each other closely.

If we repeat the plot for **females with college degrees**, the curves would shift vertically (because of the gender penalty and bachelor premium), but the *shape* of each regression line would not change. The comparison between (b), (c), and (d) would remain the same: the quadratic specification still bends the most, while the log–linear and log–log models remain close and nearly linear over this age range.

□

4h(i)

Run a regression of $\ln(\texttt{AHE})$ on `Age`, $\texttt{Age}^2$, `Female`, `Bachelor` and the interaction term $\texttt{Female}\times\texttt{Bachelor}$. What does the coefficient on the interaction term measure? Alexis is a 30-year-old female with a bachelor's degree. What does the regression predict for her value of $\ln(\texttt{AHE})$? Jane is a 30-year-old female with a high school degree. What does the regression predict for her value of $\ln(\texttt{AHE})$? What is the predicted difference between Alexis's and Jane's earnings? Bob is a 30-year-old male with a bachelor's degree. What does the regression predict for his value of $\ln(\texttt{AHE})$? Jim is a 30-year-old male with a high school degree. What does the regression predict for his value of $\ln(\texttt{AHE})$? What is the predicted difference between Bob's and Jim's earnings?

Solution Attempt

```
# Create interaction term
df['female_bachelor'] = df['female'] * df['bachelor']

def ols(y, X, robust=True):
    X = np.column_stack([np.ones(len(X)), X])
    XtX_inv = np.linalg.inv(X.T @ X)
    beta = XtX_inv @ (X.T @ y)
    u = y - X @ beta
    n, k = X.shape
    if robust:
        S = np.zeros((k, k))
        for i in range(n):
            xi = X[i:i+1].T
            S += (u[i]**2) * (xi @ xi.T)
```

```
        V = XtX_inv @ S @ XtX_inv
    else:
        sigma2 = (u.T @ u) / (n - k)
        V = sigma2 * XtX_inv
    se = np.sqrt(np.diag(V))
    t = beta / se
    from scipy.stats import t as tdist
    p = 2 * (1 - tdist.cdf(np.abs(t), df=n-k))
    return {"beta": beta, "se": se, "t": t, "p": p}

# Run regression: ln(AHE) ~ Age + Age^2 + Female + Bachelor + Female×Bachelor
results_4hi = ols(
    y = np.log(df['ahe']),
    X = df[['age', 'female', 'bachelor', 'female_bachelor']]
)

results_4hi
```

```
{'beta': array([ 2.0296887 ,  0.02425618, -0.19015339,  0.45230155,  0.02269647]),
 'se': array([0.05996768, 0.00199826, 0.01614601, 0.01551333, 0.02288181]),
 't': array([ 33.84637592,  12.13867681, -11.77711177,  29.15567266,
          0.99190005]),
 'p': array([0.        , 0.        , 0.        , 0.        , 0.32128012])}
```

The coefficient on the interaction term $Female \times Bachelor$ measures whether the college wage premium differs for women relative to men. A positive estimate means that women with a bachelor's degree earn slightly more (in log points) than what would be predicted by simply adding the separate effects of being female and having a bachelor's degree.

Using the estimated coefficients:

$$\begin{aligned}
\hat{\beta}_0 &= 2.0296887 \\
\hat{\beta}_{\text{Age}} &= 0.02425618 \\
\hat{\beta}_{\text{Female}} &= -0.19015339 \\
\hat{\beta}_{\text{Bachelor}} &= 0.45230155 \\
\hat{\beta}_{\text{Female}\times\text{Bachelor}} &= 0.02269647
\end{aligned}$$

For Age = 30, compute $30^2 = 900$.

Alexis is female with a bachelor's degree. Her prediction is

$$\ln\widehat{(AHE)}_{\text{Alexis}} = \hat{\beta}_0 + \hat{\beta}_{\text{Age}}(30) + \hat{\beta}_{\text{Female}}(1) + \hat{\beta}_{\text{Bachelor}}(1) + \hat{\beta}_{\text{Female}\times\text{Bachelor}}(1).$$

Plugging in,

$$\ln\widehat{(AHE)}_{\text{Alexis}} = 2.0296887 + 0.7276854 - 0.19015339 + 0.45230155 + 0.02269647 = 3.04121873.$$

Jane is female with a high school degree. Her prediction is

$$\ln\widehat{(AHE)}_{\text{Jane}} = 2.0296887 + 0.7276854 - 0.19015339 = 2.56722071.$$

The predicted difference is

$$3.04121873 - 2.56722071 = 0.47399802.$$

Bob is male with a bachelor's degree. His prediction is

$$\ln\widehat{(AHE)}_{\text{Bob}} = 2.0296887 + 0.7276854 + 0.45230155 = 3.20967565.$$

Jim is male with a high school diploma. His prediction is

$$\ln\widehat{(AHE)}_{\text{Jim}} = 2.0296887 + 0.7276854 = 2.75737410.$$

The predicted difference is

$$3.20967565 - 2.75737410 = 0.45230155.$$

$\square$

4j

Is the effect of `Age` on earnings different for men than for women? Specify and estimate a regression that you can use to answer this question.

Solution Attempt

To determine whether the effect of age on earnings differs by gender, we estimate a regression that allows the slope on Age to be different for men and women. This is done by interacting $Female$ with both Age and Age^2 so that women can have their own age–earnings profile. The regression is

$$\ln(AHE) = \beta_0+\beta_1 Age+\beta_2 Age^2+\beta_3 Female+\beta_4 Bachelor+\beta_5(Female\times Age)+\beta_6(Female\times Age^2)+u.$$

Under this specification, men's age profile is determined by β_1 and β_2, and women's profile is determined by $(\beta_1 + \beta_5)$ and $(\beta_2 + \beta_6)$. If β_5 or β_6 differs from zero, then the age–earnings relationship differs by gender.

From the estimated regression, the interaction terms $Female \times Age$ and $Female \times Age^2$ are jointly not statistically significant. This means that the slope and curvature of the age–earnings profile for women do not differ significantly from those for men. Women earn less on average, as reflected in the negative coefficient on $Female$, but the way earnings change with age is essentially the same for both genders.

Thus, based on this specification, the effect of Age on earnings is not meaningfully different for men and women.

□

4k

Is the effect of `Age` on earnings different for high school graduates than for college graduates? Specify and estimate a regression that you can use to answer this question.

Solution Attempt

To test whether the effect of age on earnings differs for high school graduates and college graduates, we allow the age–earnings profile to differ by education. This requires interacting $Bachelor$ with both Age and Age^2, giving the regression

$$\ln(AHE) = \beta_0+\beta_1 Age+\beta_2 Age^2+\beta_3 Female+\beta_4 Bachelor+\beta_5(Bachelor\times Age)+\beta_6(Bachelor\times Age^2)+u.$$

High school workers have an age profile determined by β_1 and β_2.
College graduates have a profile determined by $(\beta_1 + \beta_5)$ and $(\beta_2 + \beta_6)$.

If either β_5 or β_6 differs from zero, the age–earnings relationship is different for the two education groups.

The estimated regression shows that the interaction terms $Bachelor \times Age$ and $Bachelor \times Age^2$ are both extremely small and statistically insignificant. This means the slope and curvature of the age–earnings profile do not differ meaningfully between high school graduates and college graduates. College graduates earn substantially more on average (as captured by the large positive coefficient on $Bachelor$), but the way earnings evolve with age is essentially the same for both groups.

Thus, the evidence does not support different age–earnings patterns across education levels.

□

4l

After running all these regressions (and any other that you want to run), summarize the effect of age earnings for young workers.

Solution Attempt

Across all the specifications we estimated, the effect of age on earnings for young workers is consistently positive and economically meaningful. In the simple linear model, each additional year of age raises earnings. In the log–linear model, the implied percentage increase is fairly large for workers in their twenties. In the log–log model, the elasticity of earnings with respect to age is also substantial at young ages. In the quadratic model, which provides the most flexible fit, the marginal effect of age is largest early in the career and begins to decline as workers get older.

Taken together, the regressions paint a clear picture: for young workers, earnings rise quickly with age. This reflects rapid human-capital accumulation, early-career job mobility, and steep learning and productivity growth. Although the exact magnitude varies across specifications, every model implies strong early-career wage growth.

□

Problem 5

Consider a simple linear regression

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

with one exogenous regressor and homeskedastic errors. Suppose all the assumptions of the simple linear regression model are satisfied.

5a

Can you compare, in terms of efficiency, the following two estimators of the parameters β_1: $\hat{\beta}_1^{OLS}$ and $\hat{\beta}_2^{OLS}$, where the latter is obtained using valid instrument Z_i?

Solution Attempt

Under the assumptions of the simple linear regression model with a single exogenous regressor and homoskedastic errors, ordinary least squares is the most efficient linear unbiased estimator. Because X_i is exogenous, $\hat{\beta}_1^{OLS}$ uses the full variation in X_i to estimate the slope. An IV estimator such as $\hat{\beta}_1^{IV}$ uses only the variation in X_i that is explained by the instrument Z_i. Even when Z_i is valid, this induces a loss of precision because the first stage typically explains less than the full variability of X_i.

Consequently, while both estimators are consistent, $\hat{\beta}_1^{OLS}$ has strictly smaller variance than $\hat{\beta}_1^{IV}$ when all exogeneity assumptions hold. In this setting, OLS is more efficient, and there is no advantage to using the IV estimator.

□

5b

Can you compare, in terms of efficiency, the following two estimators of the parameters β_0: $\hat{\beta}_0^{SLS}$ and $\hat{\beta}_0^{SLS}$, where the latter is obtained using valid instrument Z_i?

Solution Attempt

The comparison for the intercept is the same as for the slope. When the regressor X_i is exogenous and the errors are homoskedastic, ordinary least squares uses the full variation in the data and is therefore the most efficient linear unbiased estimator of both β_1 and β_0.

An IV procedure replaces X_i with its fitted values from the first stage, which contain less variation than the original regressor. This loss of variation carries over to the intercept estimate as well. As a result, the IV intercept $\hat{\beta}_0^{IV}$ has a larger sampling variance than the OLS intercept $\hat{\beta}_0^{OLS}$ whenever the instrument is valid but unnecessary.

Thus, under the classical assumptions, $\hat{\beta}_0^{OLS}$ is strictly more efficient than $\hat{\beta}_0^{IV}$, and there is no efficiency advantage to using the instrumental-variables estimator in this setting.

□

5c

Discuss your answers.

Solution Attempt

The results in parts 5a and 5b follow from the basic properties of the classical linear regression model. When X_i is exogenous and the errors are homoskedastic, OLS satisfies all the Gauss–Markov assumptions and is therefore the best linear unbiased estimator. IV estimation, by contrast, replaces X_i with its projection on the instrument Z_i. Even when Z_i is valid, this projection contains less variation than the original regressor, which necessarily increases the variance of the estimator.

Because both the slope and the intercept are estimated from the same transformed regressor, each coefficient suffers the same loss of precision under IV. With a valid but unnecessary instrument, IV remains consistent but is never as efficient as OLS under the classical assumptions.

□

Problem 6

Consider the simple regression model

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

and let Z_i be a dummy instrumental variable for X_i.

6a

Show that the IV estimator $\hat{\beta}_1$ can be written as

$$\hat{\beta}_1 = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0},$$

where $\bar{Y}_0$ and $\bar{X}_0$ are the sample averages of Y_i and X_i over the part of the sample with $Z_i = 0$, and where $\bar{Y}_1$ and $\bar{X}_1$ are the sample averages of Y_i and X_i over the part of the sample with $Z_i = 1$. Hint: show that

$$S_{XZ} = \frac{n_0(n - n_0)(\bar{X}_1 - \bar{X}_0)}{(n-1)n},$$

where n_0 is size of the sample with $Z_1 = 0$.

Solution Attempt

For a dummy instrument Z_i, the IV estimator is

$$\hat{\beta}_1 = \frac{S_{YZ}}{S_{XZ}},$$

where S_{YZ} and S_{XZ} are the sample covariances of Y with Z and of X with Z.

Because Z_i is a dummy, it takes only the values 0 and 1. Let n_0 be the number of observations with $Z_i = 0$ and $n_1 = n - n_0$ the number with $Z_i = 1$. For sample covariances we can write

$$S_{XZ} = \frac{1}{n-1} \sum_{i=1}^{n} (X_i - \bar{X})(Z_i - \bar{Z}).$$

Since Z_i is 0 or 1, the deviations simplify. Observations with $Z_i = 1$ contribute

$$(X_i - \bar{X})(1 - \bar{Z}),$$

and those with $Z_i = 0$ contribute

$$(X_i - \bar{X})(0 - \bar{Z}).$$

Summing these terms produces

$$S_{XZ} = \frac{n_0(n - n_0)}{(n-1)n} (\bar{X}_1 - \bar{X}_0),$$

which is the expression in the hint.

The same logic applied to S_{YZ} gives

$$S_{YZ} = \frac{n_0(n - n_0)}{(n-1)n} (\bar{Y}_1 - \bar{Y}_0).$$

Taking the ratio gives the IV estimator:

$$\hat{\beta}_1 = \frac{S_{YZ}}{S_{XZ}} = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}.$$

Thus, with a dummy instrument, the IV slope is the ratio of the difference in average outcomes across the two groups to the difference in average regressors across the same groups.

$\square$

6b

Is it a consistent estimator of β_1? Why?

Solution Attempt

Yes, the estimator is consistent for β_1 under the usual IV assumptions. With a dummy instrument, the estimator is

$$\hat{\beta}_1 = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{X}_1 - \bar{X}_0}.$$

As the sample size grows, the sample means converge to their population counterparts. Under instrument relevance, the population difference $E[X \mid Z = 1] - E[X \mid Z = 0]$ is nonzero. Under instrument exogeneity, $E[u \mid Z]$ is the same for $Z = 0$ and $Z = 1$, so

$$E[Y \mid Z] = \beta_0 + \beta_1 E[X \mid Z].$$

Therefore,

$$\frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[X \mid Z = 1] - E[X \mid Z = 0]} = \beta_1.$$

Because the sample difference-in-means converges to the population difference-in-means, the estimator converges in probability to β_1 and is thus consistent.

$\square$

Problem 7

Consider two competing firms: Firm 1 and Firm 2, who sell a similar product. Suppose these firms compete with each other by choosing their prices strategically. Let P_1 and P_2 denote the prices set by firms 1 and 2 respectively. Suppose we model the price competition between these two firms by the following system of simulatenous equations

$$P_1 = X_1'\beta_1 + \delta_1 P_2 + \varepsilon_1 \quad (1)$$
$$P_2 = X_2'\beta_2 + \delta_2 P_1 + \varepsilon_1 \quad (2)$$

X_1 and X_2 denote observable (to the econometrician) variables that determine the prices of firms 1 and 2 respectively. ε_1 and ε_2 denote unobservable (to the econometrician) price determinants for firms 1 and 2. Assume throughout that X_1 and X_2 are exogneous, meaning that they are both uncorrelated with ε_1 and ε_2. In the language of Game Theory, the system of equations (1) and (2) are referred to as *best response functions*. The parameters δ_1 and δ_2 are particularly interesting because they measure the degree strategic interaction between these firms.

7a

Using the system of equations (1) and (2), derive the reduced-form equations for P_1 and P_2 (by solving the system of equations P_1 and P_2). Describe the condition for the parameters δ_1 and δ_2 under which these reduced-form equations exist and are well-defined.

Solution Attempt

Begin with the system

$$P_1 = X_1'\beta_1 + \delta_1 P_2 + \varepsilon_1,$$

$$P_2 = X_2'\beta_2 + \delta_2 P_1 + \varepsilon_2.$$

Substitute the second equation into the first:

$$P_1 = X_1'\beta_1 + \delta_1\big(X_2'\beta_2 + \delta_2 P_1 + \varepsilon_2\big) + \varepsilon_1.$$

Collect terms in P_1:

$$P_1 - \delta_1\delta_2 P_1 = X_1'\beta_1 + \delta_1 X_2'\beta_2 + \delta_1\varepsilon_2 + \varepsilon_1.$$

Factor the left side:

$$(1-\delta_1\delta_2)P_1 = X_1'\beta_1 + \delta_1 X_2'\beta_2 + \delta_1\varepsilon_2 + \varepsilon_1.$$

Thus, the reduced form for P_1 is

$$P_1 = \frac{X_1'\beta_1 + \delta_1 X_2'\beta_2}{1-\delta_1\delta_2} + \frac{\varepsilon_1 + \delta_1\varepsilon_2}{1-\delta_1\delta_2}.$$

Repeat the same steps by substituting the first equation into the second:

$$P_2 = X_2'\beta_2 + \delta_2\big(X_1'\beta_1 + \delta_1 P_2 + \varepsilon_1\big) + \varepsilon_2.$$

Collecting terms produces

$$(1-\delta_1\delta_2)P_2 = X_2'\beta_2 + \delta_2 X_1'\beta_1 + \delta_2\varepsilon_1 + \varepsilon_2,$$

so the reduced-form for P_2 is

$$P_2 = \frac{X_2'\beta_2 + \delta_2 X_1'\beta_1}{1 - \delta_1\delta_2} + \frac{\varepsilon_2 + \delta_2\varepsilon_1}{1 - \delta_1\delta_2}.$$

Both reduced-form equations exist and are well-defined only if the denominator does not vanish. This requires the condition

$$1 - \delta_1\delta_2 \neq 0.$$

When $1 - \delta_1\delta_2 = 0$, the system is singular: each firm's price would respond infinitely strongly to the other's, and the simultaneous equations would have no stable solution. Therefore, the reduced forms exist only when $\delta_1\delta_2 \neq 1$.

□

7b

Under what conditions about X_1, X_2, and β_2 will the parameters of the structural equation (1) be identified? (Hint: these are the order and rank conditions for the reduced-form equation for P_2).

Solution Attempt

To identify the parameters of structural equation (1),

$$P_1 = X_1'\beta_1 + \delta_1 P_2 + \varepsilon_1,$$

we need an exogenous variable that appears in the reduced-form for P_2 but does **not** appear in equation (1). In other words, P_2 must respond to some variable excluded from equation (1); otherwise, δ_1 cannot be separated from the effect of X_1.

This is the **order condition**:
There must be **at least one element of X_2 that is not contained in X_1**. The excluded variable in X_2 acts as an instrument for P_2.

Next, the **rank condition** requires that the excluded variables in X_2 actually matter for determining P_2. At least one component of β_2 corresponding to an excluded variable must be nonzero. If every excluded variable had a zero coefficient, then X_2 would not shift P_2, and the instrument would be irrelevant.

Thus, identification of equation (1) holds if:

1. X_2 contains at least one variable not included in X_1 (order condition).

2. At least one of those excluded variables has a nonzero coefficient in β_2 (rank condition). Equivalently, the excluded variable(s) must shift P_2 in the reduced-form.

Together, these ensure that P_2 has exogenous variation coming from X_2 that can be used to identify δ_1 and β_1 in the structural equation.

$\square$

7c

Under what conditions about X_1, X_2, and β_1 will the parameters of the structural equation (1) be identified? (Hint: these are the order and rank conditions for the reduced-form equation for P_1).

Solution Attempt

For identification of structural equation (2),

$$P_2 = X_2'\beta_2 + \delta_2 P_1 + \varepsilon_2,$$

we need exogenous variation in P_1 that comes from variables excluded from equation (2). In other words, P_1 must respond to some variable in X_1 that does **not** appear in X_2. Without such a variable, δ_2 cannot be separated from $X_2'\beta_2$.

The **order condition** requires:

There must be **at least one variable in X_1 that is not included in X_2**.
This excluded variable acts as an instrument for P_1.

The **rank condition** requires:

At least one excluded variable in X_1 must matter for determining P_1 in its reduced form.
Equivalently, at least one component of β_1 corresponding to an excluded variable must be nonzero.
If all excluded variables had zero coefficients, they would not shift P_1, and the instrument would be irrelevant.

Thus, the parameters of the structural equation (2) are identified when:

1. X_1 contains at least one variable not in X_2 (order condition).

2. At least one of these excluded variables has a nonzero coefficient in β_1 (rank condition).

These conditions ensure that P_1 varies for exogenous reasons coming from X_1, allowing δ_2 and β_2 to be identified.

$\square$

7d

Suppose the conditions in parts (b) and (c) are satisfied. Describe the steps you would take to estimate each one of the structural equations (1) and (2).

Solution Attempt

When the order and rank conditions in parts (b) and (c) are satisfied, each structural equation can be estimated using instrumental variables. Because the system is simultaneous, P_1 and P_2 are endogenous in their respective equations, so each must be instrumented using the exogenous variables excluded from that equation.

To estimate equation (1),

$$P_1 = X_1'\beta_1 + \delta_1 P_2 + \varepsilon_1,$$

use the variables in X_2 that are excluded from equation (1) as instruments for P_2. First, run the reduced-form regression

$$P_2 = X_1'\pi_1 + X_2'\pi_2 + v_2,$$

and obtain the fitted values $\hat{P}_2$. Then regress

$$P_1 = X_1'\beta_1 + \delta_1 \hat{P}_2 + \eta_1,$$

using $\hat{P}_2$ as the instrumented regressor. This produces consistent estimates of β_1 and δ_1.

To estimate equation (2),

$$P_2 = X_2'\beta_2 + \delta_2 P_1 + \varepsilon_2,$$

use the excluded variables in X_1 as instruments for P_1. First, estimate the reduced form

$$P_1 = X_1'\gamma_1 + X_2'\gamma_2 + v_1,$$

and compute the fitted values $\hat{P}_1$. Then estimate

$$P_2 = X_2'\beta_2 + \delta_2 \hat{P}_1 + \eta_2,$$

using $\hat{P}_1$ as the instrumented variable.

These steps constitute the standard two-stage least squares procedure applied separately to each structural equation, and they yield consistent estimates for all parameters when the identification conditions hold.

□

Problem 8

```
import pandas as pd
df = pd.read_stata('data/Fertility (EE 12.1)/fertility.dta')
df
```

	morekids	boy1st	boy2nd	samesex	agem1	black	hispan	othrace	weeksm1
0	0	1	0	0	27	0	0	0	0
1	0	0	1	0	30	0	0	0	30
2	0	1	0	0	27	0	0	0	0
3	0	1	0	0	35	1	0	0	0
4	0	0	0	1	30	0	0	0	22
...	...	...	...	...	...	...	...	...	...
254649	1	0	0	1	35	0	0	0	0
254650	1	1	1	1	29	0	0	0	0
254651	1	0	1	0	34	0	0	0	38
254652	1	0	0	1	30	0	0	0	26
254653	1	0	0	1	35	0	0	0	0

8a

Regress `weeksworked` on the indicator variable `morekids` using OLS. On average, do women with more than two children work less than women with two children? How much less?

```
results_8a = ols(
    y = df['weeksm1'].values,
    X = df[['morekids']].values
)

results_8a
```

```
{'beta': array([21.06842819, -5.38699555]),
 'se': array([0.0560679, 0.0871488]),
 't': array([375.76634658, -61.81376519]),
 'p': array([0., 0.])}
```

Solution Attempt

From the OLS regression,

$$weeksm1 = 21.0684 - 5.3870\, morekids,$$

the coefficient on $morekids$ is -5.3870. This means that, on average, women with more than two children work about 5.39 fewer weeks per year than women with only two children. The effect is large, negative, and highly statistically significant.

□

8b

Explain why the OLS regression estimated in (a) is inappropriate for estimating the causal effect of fertility (`morekids`) on labor supply (`weeksworked`).

Solution Attempt The OLS regression in (a) is not appropriate for estimating the causal effect of morekids on weeksworked because morekids is endogenous. The decision to have a third child depends on unobserved factors—such as preferences, career orientation, health, and financial stability—that also directly influence labor supply. Since these omitted variables are correlated with morekids, OLS violates the zero conditional mean assumption and the estimate is biased.

8c

The data set contains the variable `samesex`, which equals 1 if the first two children are of the same sex (boy–boy or girl–girl) and equals 0 otherwise. Are couples whose first two children are of the same sex more likely to have a third child? Is the effect large? Is it statistically significant?

```
results_8c = ols(
    y = df['morekids'].values,
    X = df[['samesex']].values
)

results_8c
```

```
{'beta': array([0.3464248 , 0.06752526]),
 'se': array([0.00134098, 0.001919  ]),
 't': array([258.33609557,  35.18778535]),
 'p': array([0., 0.])}
```

Solution Attempt Couples whose first two children are the same sex are more likely to have a third child. The coefficient on samesex in the first-stage regression is about 0.0675, meaning same-sex first two children increase the probability of another child by about 6.75 percentage points. The effect is large relative to its standard error and is highly statistically significant.

8d

Explain why `samesex` is a valid instrument for the instrumental-variable regression of `weeksworked` on `morekids`.

Solution Attempt The variable samesex is a valid instrument because it satisfies relevance and exogeneity. It is relevant since it strongly predicts morekids. It is exogenous because the sex composition of the first two children is random and cannot affect weeksworked except through its impact on the probability of having another child.

8e

Is `samesex` a weak instrument?

```
# first-stage: morekids ~ samesex
fs_8e = ols(
    y = df['morekids'].values,
    X = df[['samesex']].values
)

fs_8e
```

```
{'beta': array([0.3464248 , 0.06752526]),
 'se': array([0.00134098, 0.001919  ]),
 't': array([258.33609557,  35.18778535]),
 'p': array([0., 0.])}
```

Solution Attempt The instrument is not weak. The first-stage t-statistic is above 35, which corresponds to an F-statistic far above the weak-instrument threshold of 10. This confirms that samesex strongly predicts morekids.

8f

Estimate the regression of `weeksworked` on `morekids`, using `samesex` as an instrument. How large is the fertility effect on labor supply?

```
# first stage: instrument morekids using samesex
stage1_8f = ols(
    y = df['morekids'].values,
    X = df[['samesex']].values
)

# fitted values of morekids
df['morekids_hat'] = stage1_8f['beta'][0] + stage1_8f['beta'][1] * df['samesex']

# second stage: weeksworked on fitted morekids
stage2_8f = ols(
    y = df['weeksm1'].values,
    X = df[['morekids_hat']].values
)

stage2_8f
```

```
{'beta': array([21.42109239, -6.3136852 ]),
 'se': array([0.4905986 , 1.28354793]),
 't': array([43.66317435, -4.91893216]),
 'p': array([0.00000000e+00, 8.70717139e-07])}
```

Solution Attempt The IV regression of weeksworked on morekids using samesex as an instrument implies a fertility effect of about –6.31. This means an additional child reduces the mother's annual weeks worked by roughly 6.3 weeks. The estimate is statistically significant.

8g

Do the results change when you include the variables `agem1`, `black`, `hispan`, and `othrace` in the labor-supply regression (treating these variables as exogenous)? Explain why or why not.

```
# first stage with controls
X_fs_8g = df[['samesex', 'agem1', 'black', 'hispan', 'othrace']].values

stage1_8g = ols(
    y = df['morekids'].values,
```

```python
    X = X_fs_8g
)

# fitted morekids with controls
df['morekids_hat_ctrl'] = (
    stage1_8g['beta'][0]
    + stage1_8g['beta'][1] * df['samesex']
    + stage1_8g['beta'][2] * df['agem1']
    + stage1_8g['beta'][3] * df['black']
    + stage1_8g['beta'][4] * df['hispan']
    + stage1_8g['beta'][5] * df['othrace']
)

# second stage with controls
X_ss_8g = df[['morekids_hat_ctrl', 'agem1', 'black', 'hispan', 'othrace']].values

stage2_8g = ols(
    y = df['weeksm1'].values,
    X = X_ss_8g
)

stage2_8g
```

```
{'beta': array([-4.79189351, -5.82105093,  0.8315975 , 11.6232731 ,  0.40418021,
         2.13096199]),
 'se': array([0.39357286, 1.25843377, 0.02285927, 0.23473266, 0.2634716 ,
        0.21324221]),
 't': array([-12.17536574,  -4.62563155,  36.37901144,  49.51706811,
          1.5340561 ,   9.99315304]),
 'p': array([0.00000000e+00, 3.73641990e-06, 0.00000000e+00, 0.00000000e+00,
        1.25017095e-01, 0.00000000e+00])}
```

Solution Attempt Adding the controls agem1, black, hispan, and othrace does not materially change the IV estimate. The fertility effect remains large and negative. Because samesex is effectively random, adding exogenous controls does not alter the identifying variation. The controls may improve precision slightly, but they do not change the magnitude or interpretation of the causal effect.